

Discuss Agenda- incl **Intro, Langflow**

Disadv of agentic ai> mcp (disadv)(nov 2024)> a2a (april 2025)

Agentic AI is a type of artificial intelligence that empowers systems to make decisions and perform tasks autonomously, without direct human intervention. These systems are capable of adapting to changing environments and learning from their actions, essentially acting as intelligent agents within a system.

Travel Plan, Software- Jira

Multiagent system

Agents vs tools

This doc content

Walk through documentation-

> **Intro**

- 1st
- 2nd: **Code**- event happening currently,

> **Concepts**

- Multimodal Agents: **Code**- (Multi-agent (**finance**))
- Teams (content In this doc, with trip eg)
- Models
- Tools: **Code**- (db.py)
- Remaining like embeddings etc

Agno Framework or Phidata

Agno is a lightweight library for building Multimodal Agents (support for text, image, audio and video) with memory, knowledge, tools and reasoning.

Here,

memory refers to the agent's ability to retain and recall information over time. This is a critical feature for creating intelligent agents that can engage in meaningful, context-aware interactions.

Agents can store **knowledge** in a vector database and use it for RAG or dynamic few-shot learning. Agno agents use Agentic RAG by default, which means they will search their knowledge base for the specific information they need to achieve their task.

In the context of Agno, **tools** refer to external services that the agent can use to perform specific tasks. These tools extend the capabilities of the agent by allowing it to interact with external systems, APIs, or resources. Tools are essential for enabling the agent to perform actions beyond its core processing abilities, such as retrieving real-time information, performing computations. One example of a tool is the DuckDuckGo Search Tool, which allows the agent to perform web searches using the DuckDuckGo search engine.

Key-features

- Run anywhere, Agno is open-source.
- Lightning Fast: Agent creation is ~10,000x faster than LangGraph
- It can build multiagents. Multi-agent systems involve creating a group of AI agents, each with a specific role or expertise, that collaborate to achieve a common goal.

Example: Multi-Agent System for Software Development

Project Manager Agent, UI Designer Agent, Backend Developer Agent, QA Tester Agent.

- Agno uses 50x less memory (~3.75 KiB) than LangGraph.

LangGraph consumes significantly more memory, which can be costly at scale.

Purpose:

- Agno is built for **high-performance**, large-scale systems (**thousands of agents**). LangGraph is more **general-purpose** and flexible but less optimized for performance.

What are Agents?

Agents are AI programs that execute tasks autonomously. They solve problems by running tools, accessing knowledge and memory to improve responses. Unlike traditional programs that follow a predefined execution path, agents dynamically adapt their approach based on context, knowledge and tool results.

This framework classifies agents based on their agency (ability to act) and autonomy (ability to make decisions independently). Instead of a binary distinction (agent vs. non-agent), it provides a graded approach to understanding the capabilities of agents, ranging from simple inference tasks to advanced collaborative systems.

Level 0: Agents with No Tools (Basic Inference Tasks)

Agency: Minimal.

Autonomy: None.

Description: These agents are limited to performing basic tasks, such as answering questions or generating text, based on their pre-trained knowledge. They cannot interact with external systems or tools.

Example: A simple chatbot that responds to user queries using a pre-trained language model (e.g., GPT-3) but cannot access the internet or perform actions.

Level 1: Agents with Tools for Autonomous Task Execution

Agency: Moderate.

Autonomy: Basic.

Description: These agents can use tools to perform tasks beyond basic inference. Tools allow them to interact with external systems, APIs, or databases, enabling them to execute actions autonomously.

Example: An agent that uses a search tool (e.g., DuckDuckGo) to retrieve real-time information or a calculator tool to perform mathematical computations.

Level 2: Agents with Knowledge (Memory and Reasoning)

Agency: High.

Autonomy: Advanced.

Description: These agents combine tools with memory and reasoning capabilities. They can retain information from past interactions, learn from experience, and make context-aware decisions.

Example: A virtual assistant that remembers user preferences, stores conversation history, and uses reasoning to provide personalized recommendations.

Level 3: Teams of Specialized Agents Collaborating on Complex Workflows

Agency: Very High.

Autonomy: Very High.

Description: At this level, multiple specialized agents work together as a team, each contributing unique skills or expertise. These agents collaborate to solve complex problems that require diverse capabilities.

Example: A team of agents working on a software development project: One agent handles code generation. Another agent performs testing and debugging. A third agent manages project documentation. A fourth agent coordinates communication between team members.

Difference between Agno/Phidata and Langgraph

Speed:

Agno is 10,000x faster at starting agents (~2μs) compared to LangGraph.

LangGraph takes much longer to initialize agents.

Memory Usage:

Agno uses 50x less memory (~3.75 KiB) than LangGraph.

LangGraph consumes significantly more memory, which can be costly at scale.

Purpose:

Agno is built for high-performance, large-scale systems (**thousands of agents**).

LangGraph is more general-purpose and flexible but less optimized for performance.

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Message
Mock drill in bangalore today?

Tool Calls
• duckduckgo_search(query=mock drill in Bangalore today, max_results=5)

Response (3.3s)
Mock Drill in Bangalore Today

To get the latest information on a mock drill in Bangalore today, I will search the web for the most recent updates.

Mock Drill in Bangalore Today

There is a mock drill scheduled in Bangalore today, May 7, 2025, at 4 PM, as part of Operation Abhyaas, a civil defence mock drill. The drill aims to test emergency response and raise public awareness on safety during crises.

The mock drill will take place at 32 locations in Bengaluru, including high-alert zones, and will simulate air raids and blackouts. The city is equipped with 35 air raid sirens, of which 32 are functional.

Please note that the mock drill is a nationwide exercise, and cities like Mumbai, Delhi, and Chennai are also participating.

It is essential to be aware of the mock drill and to follow the instructions provided by the authorities to ensure public safety.

Key Points:

• Mock drill scheduled for May 7, 2025, at 4 PM
• 32 locations in Bengaluru will participate in the drill
• The drill will simulate air raids and blackouts
• 35 air raid sirens are installed in the city, with 32 being functional
• The exercise is part of Operation Abhyaas, a civil defence mock drill
• The drill aims to test emergency response and raise public awareness on safety during crises
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Response (20.9s)
Market Outlook and Financial Performance Analysis of NVDA and Tesla

The market outlook for NVDA and Tesla is complex, with various factors influencing their stock prices. Based on the search results, here is a summary of their financial performance and market trends:

NVDA (NVIDIA Corporation)

• The company's revenue was $130.50 billion in 2024, representing a 114.20% increase from the previous year.
• Earnings were $72.88 billion, a 144.89% increase from the previous year.
• The total return for NVDA stock is 33.47% over the past 12 months, outperforming the S&P 500.
• The stock price has been affected by the company's efforts to curb chip sales to China, with a 6% decline in stock price.

Tesla (TSLA)

• The company's revenue was $97.69 billion in 2024, a 0.95% increase from the previous year.
• The stock price has been volatile, with a significant decline from its all-time high in 2025.
• Tesla's electric vehicle business has been impacted by various factors, including government regulations and competition in the industry.
• The company's AI updates have also affected its stock price, with some investors expressing concerns about the company's focus on AI development.

Comparison of NVDA and Tesla

• Both companies are considered disruptive technology companies, with a strong presence in their respective markets.
• NVDA has outperformed Tesla in terms of revenue growth and stock price performance over the past 12 months.
• However, Tesla's electric vehicle business has significant growth potential, and the company's AI updates could provide a competitive advantage in the industry.

Recommendations

Based on the analysis, it is recommended that investors consider the following:

• NVDA: The company's strong financial performance and outperformance of the S&P 500 make it a promising investment opportunity. However, investors should be cautious about the potential risks associated with the company's efforts to curb chip sales to China.
• Tesla: The company's volatile stock price and concerns about its focus on AI development make it a higher-risk investment opportunity. However, the company's electric vehicle business has significant growth potential, and investors who are willing to take on more risk may find Tesla to be a promising investment opportunity.

Ultimately, the decision to invest in NVDA or Tesla depends on an individual's investment goals, risk tolerance, and overall financial situation. It is essential to conduct thorough research and consult with a financial advisor before making any investment decisions.
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What Are Agent Teams?

Agent Teams are groups of AI agents (or sub-teams) designed to work together to solve complex tasks more effectively than a single agent could alone. These teams can operate in different modes depending on the task requirements:

1. **Route Mode** (Task Delegation)

- The **Team Leader** acts as a **router**, analyzing the user's request and directing it to the **most suitable agent** in the team.
- Example:
 - A customer support team where the leader routes technical questions to a "Tech Support Agent" and billing questions to a "Finance Agent."
- Best for:
 - Tasks where different agents specialize in distinct subtasks.

2. **Coordinate Mode** (Sequential Workflow)

- The **Team Leader breaks down the task**, assigns subtasks to different agents, and **combines their outputs** into a final response.
- Example:
 - A research team where one agent gathers data, another analyzes it, and the leader compiles the report.
- Best for:
 - Multi-step workflows where agents contribute sequentially.

3. **Collaborate Mode** (Parallel Workflow)

- **All agents work on the same task simultaneously**, and the leader **synthesizes their responses** into a cohesive answer.
- Example:
 - A brainstorming session where multiple agents generate ideas, and the leader picks the best ones.
- Best for:
 - Tasks requiring diverse perspectives or redundancy for accuracy.

"Identify the language of the user's question and direct it to the appropriate language agent.",

"If the user asks in a language whose agent is not a team member, respond in English with:",

"I can only answer in the following languages: English, Chinese, French. Please ask your question in one of these languages.",

"Always check the language of the user's input before routing to an agent.",

"For unsupported languages like Italian, respond in English with the above message.",

Reasoning Models

Reasoning models are a new class of large language models trained with reinforcement learning to think before they answer. They produce a long internal chain of thought before responding. Examples of reasoning models include:

OpenAI o1-pro and o3-mini

Claude 3.7 sonnet in extended-thinking mode

Gemini 2.0 flash thinking

DeepSeek-R1

Reasoning models deeply consider and think through a plan before taking action.

Its all about what the model does before it starts generating a response.

Reasoning models excel at single-shot use-cases. They're perfect for solving hard problems (coding, math, physics) that don't require multiple turns, or calling tools sequentially.

Differences between knowledge base and database

Attribute	Knowledge Base	Database
Usage	Knowledge bases are used to store and share knowledge and expertise and are focused on making information easy to understand.	Databases are used to store and organize data and are primarily concerned with accuracy.
Structure	Knowledge bases are often more loosely structured and designed to optimize search results.	Databases are structured in a way that allows for efficient data storage and retrieval.
Accessibility	Knowledge bases are usually open and accessible to a wider audience.	Databases are often used to store transactional data and are not accessible to the public.
Storage	Knowledge bases are used to store knowledge and expertise.	Databases are used to store data.

What is a Database?

A [database](#) is an organized collection of data. It stores information in a structured format that is easy to access and update and is generally used for internal data storage and retrieval. Common examples of databases include customer information, product inventory management, and financial records.

What is a Knowledge base?

A [knowledge base](#), on the other hand, is an organized collection of information related to a specific topic. Knowledge bases are typically created for external use,

providing customers and other stakeholders with quick access to information.
Knowledge bases often contain FAQs, user guides, and other resources.

Data Management

A knowledge base stores text, images, and audio information, to provide answers to customer questions, additional product information, and other resources. A database, on the other hand, is designed to store structured data such as numerical values.